

Contents

Symbols	3
1 Introduction	5
1.1 Motivation	5
1.2 Dissertation Outline	8
2 Theory	9
2.1 Hypersonic overview	9
2.2 Turbulence overview	11
2.3 Optics overview	11
3 Modeling	12
4 Numerical Results	13
Appendices	16

List of Tables

1.1	Schematic of the fidelity of aero-optical simulations.	8
-----	--	---

Symbols

List of Figures

1.1 Optics Schematic	7
2.1 Additional Degrees of Freedom	10

Chapter 1

Introduction

1.1 Motivation

Several reports in the classified and unclassified [1], [2] environment have stated the concern of aero-optic effects at hypersonic speeds to be of paramount importance for national security. Applied work in this field is not only being done by national laboratories (Lawrence Livermore National Laboratory, Applied Physics Laboratory) but also by the Department of Defense (DoD) contractors (Lockheed Martin Corp., Northrop Grumman Corp., Raytheon Technologies); as expected most of this work is limited in the open literature. However, work that investigates the effect of optical properties in a hypersonic environment is lacking and there are many fundamental phenomena such as index of refraction, that needs to be better understood.

Understanding the effects of the environment in which an optical beam traverse is imperative in target detection. Radar engineers will be mainly concerned with the atmospheric environment (rain, clouds, fog, snow) but will not much much consideration on the impact aerodynamic effects such as shock wave and boundary layer interactions has on the optical beam. Some work showing how much the index of refraction is affected by aerodynamic effects has been performed by Gupta et.al. [3] for supersonic flows and showed that there is indeed, a change on index of refraction

across a shock wave. For subsonic flows, it might be a valid assumption to ignore the aerodynamic effects on the optical beam; however for supersonic and specially for hypersonic flows this assumption has to be revisited. During a hypersonic flight the temperatures behind the shock are very high, that will not only induce thermal noise on the sensor but will also affect the electromagnetic properties that will influence the sensor's performance. Hence a higher distortion as well as power loss will be expected in a hypersonic flow. Being able to accurately understand this is of importance to build accurate sensor that can not only target objects moving at these speeds, but can also accurately predict targets from an object moving at hypersonic speeds.

Fig.1.1, shows the path that a beam in a hypersonic flight will follow. The beam has to travel across the isolated gas, the radome, the boundary layer, shock layer and lastly environment. Traditionally, radar engineers ignore aerodynamic effects (shock waves, expansion waves) for their analysis; however for the case of hypersonic flights the aerodynamic effects cannot be ignored anymore, because the electromagnetic properties across the boundary layer and shock layer change.

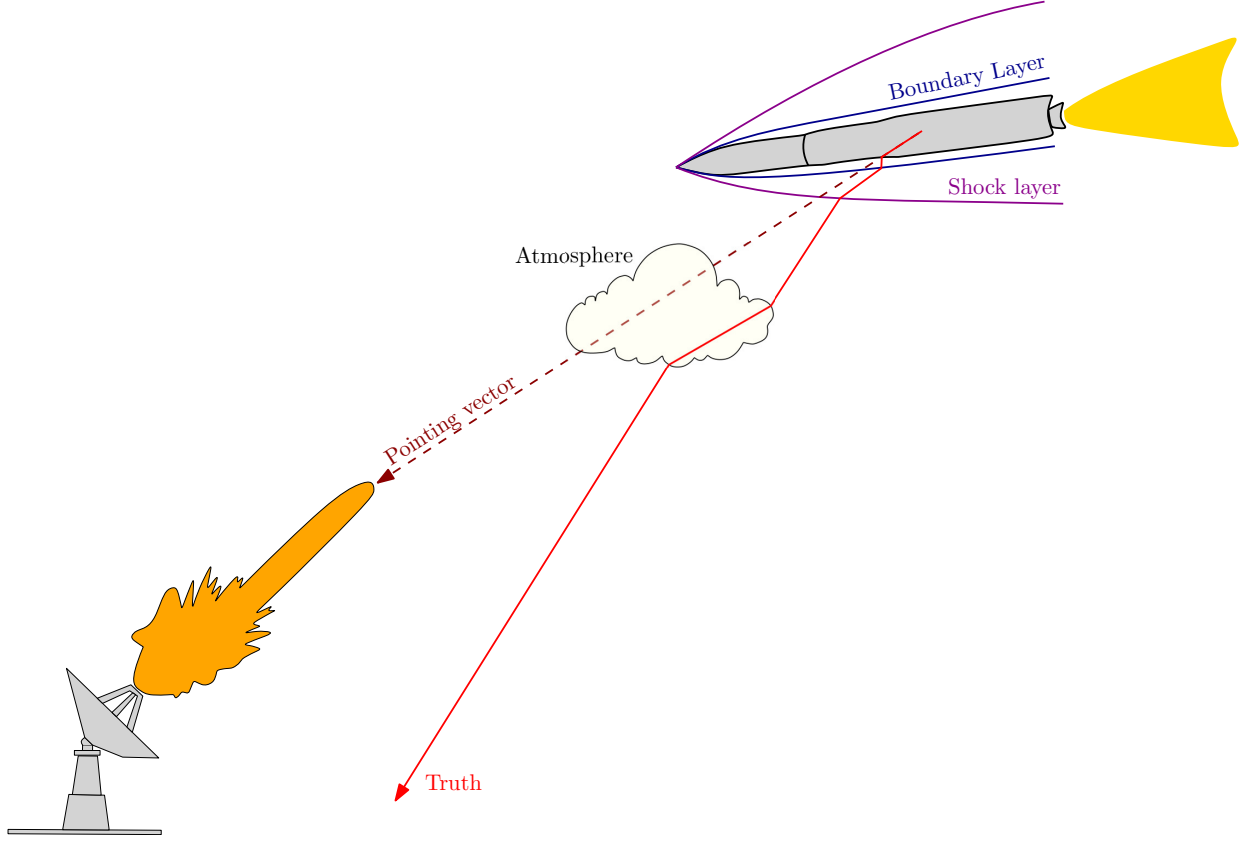


Figure 1.1: Optics Schematic

Aero-optical investigations can be split into four quadrants as shown in Tab.1.1. The optical parameters are contingent on the degree to which turbulence eddy resolution and turbulence fluctuations are captured in the Computational Fluid Dynamics (CFD) simulation. In cases where a lower level of flow resolution is utilized, either a laminar flow or a Reynolds Averaged Navier Stokes (RANS) simulation are employed as shown in Quadrants I and III of Tab.1.1. While a RANS approach averages out turbulent fluctuations, it is a popular approach for hypersonic vehicle analysis due to its efficiency [4] and can still provide aero-optical quantities such as Gladstone-Dale constant, polarizability, boresight error (BSE), Optical Path Length (OPL), and wavefront distortion (WD). The amount of enthalpy is also an important criteria in the flow field. With high enthalpy flows, many physical phenomena in the flow field that are not present in low-speed flows present themselves. This includes nonequilibrium, reactive flows, and excitation of molecular internal modes, all of which can affect optical signals either directly or indirectly; these are shown in the top Quadrants.

This work aims to investigate and compare all quadrants using different fidelities of nonequilibrium and fluid flows.

Quadrant III: High-Enthalpy, Low-Flow Resolution <i>Physics captured:</i> chemical and thermal nonequilibrium, shock structures, bulk unsteadiness, bulk density gradients. <i>AO quantities measured:</i> Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD.	Quadrant IV: High-Enthalpy, High-Flow Resolution <i>Physics captured:</i> chemical and thermal nonequilibrium, shock structures, bulk unsteadiness, bulk density gradients. <i>AO quantities measured:</i> Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD, OPD.
Quadrant I: Low-Enthalpy, Low-Flow Resolution <i>Physics captured:</i> shock structures, bulk unsteadiness, bulk density gradients. <i>AO quantities measured:</i> Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD.	Quadrant II: Low-Enthalpy, High-Flow Resolution <i>Physics captured:</i> shock structures, bulk unsteadiness, bulk density gradients. <i>AO quantities measured:</i> Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD, OPD.

Table 1.1: Schematic of the fidelity of aero-optical simulations.

A conference paper focusing on quadrants *I* and *III* was published by Liza et al. [5]. In this paper, the authors examined the impact of high temperatures and nonequilibrium conditions on polarizability. Additionally, they investigated the effects of Mach number, turbulence, and chemistry modeling on optical properties, including the index of refraction and the dielectric constant of air.

1.2 Dissertation Outline

Chapter 2

Theory

2.1 Hypersonic overview

The first hypersonic flight took place during the early stages of the Cold War, in 1949 when the German V2 rocket developed during the Second World War in Nazi Germany and brought to the USA (possible during operation Paperclip) reached hypersonic speeds as it entered the atmosphere [6]. In 1961, the Soviet Union cosmonaut Yuri Gagarin became the first person to safely reach space and orbit Earth. During his journey he became the first human in history to travel at hypersonic speeds [6].

Speeds in a hypersonic flow are greater than Mach five, at these speeds the kinetic energy of the gas molecules is converted into thermal energy, leading to very high temperatures. At these speeds the chemical composition of the gas becomes important. The high temperature can cause dissociation (breaking apart of molecules), and ionization (formation of charged particles) of the air molecules. At temperatures above 2000 [K] O_2 starts to dissociate, and it is completely dissociated above 4000 [K]. N_2 begins to dissociate above 4000 [K] and it is completely dissociated above 9000 [K] [7].

At these temperatures, the Hamiltonian (kinetic and potential energy) of the system has to be modified because more energy modes are accessible (rotational,

vibrational, and electronic energy). Fig.2.1 shows the additional degrees of freedom that emerge during a hypersonic flow.

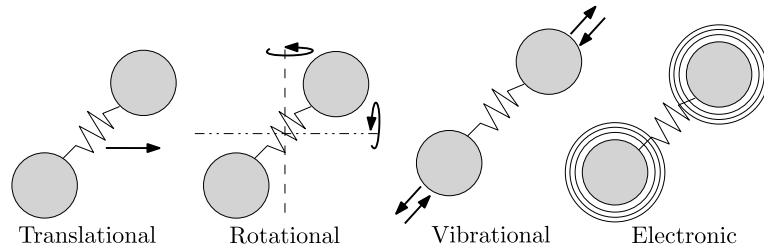


Figure 2.1: Additional Degrees of Freedom

In addition, each new accessible energy mode acts at different time scales, Eq.(2.1) show these time scales.

$$\tau_{flow} \approx \tau_{tr} < \tau_{rot} < \tau_{vib} < \tau_{chem} \quad (2.1)$$

The different time scales is the reason why flows at hypersonic speeds are in nonequilibrium. After air molecules travel across a shock, it will take certain number of collisions to convert the energy into thermal energy. In the case of translational energy is almost instantaneous, in the case of rotational is approximately 10 collisions and in the case of vibrational is approximately 1000 collisions [8]. The nonequilibrium characteristics of the flow can be characterize by comparing the flow characteristic time with the other time scales [9].

These additional energy modes increase the degrees of freedom (dof) of the system, allowing molecules to occupy more energy levels (rotational, vibrational, electronic). Due to this, the partition function (which indicates how particles in an assembly are partitioned among various energy levels of an atom or molecule [10]) has to be modified to consider these additional dofs. In statistical mechanics, the partition function is used to calculate the heat capacities of the system, which are then used to calculate macro properties such as heat transfer and temperature, which are employed in fluid dynamics.

At these conditions, the calorically perfect gas approximation is no longer suitable because this assumption does not consider thermochemical effects. The calorically perfect gas assumption will lead to errors such as over-predicting temperature, under-predicting density, and underestimating pressure. Therefore, at these speeds, thermochemical effects need to be considered and accurate models that consider these effects need to be included.

2.2 Turbulence overview

2.3 Optics overview

Chapter 3

Modeling

Chapter 4

Numerical Results

Bibliography

- [1] C. H. Jones, “Recommendations from the workshop on communications through plasma during hypersonic flight,” Tech. Rep., 2009. DOI: [10.2514/6.2009-1718](https://doi.org/10.2514/6.2009-1718).
- [2] Chief Scientist (Air Force), “Technology Horizons A Vision for Air Force Science and Technology 2010-30,” United States Air Force Chief Scientist, Tech. Rep., 2010.
- [3] A. Gupta and B. Argrow, “Analytical Approach for Aero-Optical and Atmospheric Effects in Supersonic Flow Fields,” in *AIAA Scitech 2020 Forum*, Reston, Virginia: American Institute of Aeronautics and Astronautics, Jan. 2020, ISBN: 978-1-62410-595-1. DOI: [10.2514/6.2020-0684](https://doi.org/10.2514/6.2020-0684). [Online]. Available: <https://arc.aiaa.org/doi/10.2514/6.2020-0684>.
- [4] G. V. Candler, “Rate Effects in Hypersonic Flows,” *Annual Review of Fluid Mechanics*, vol. 51, pp. 379–402, 2019, ISSN: 00664189. DOI: [10.1146/annurev-fluid-010518-040258](https://doi.org/10.1146/annurev-fluid-010518-040258).
- [5] M. Liza, O. Tumuklu, and K. M. Hanquist, “Nonequilibrium Effects on Aero-Optics in Hypersonic Flows,” in *AIAA AVIATION 2023 Forum*, Reston, Virginia: American Institute of Aeronautics and Astronautics, Jun. 2023, ISBN: 978-1-62410-704-7. DOI: [10.2514/6.2023-3736](https://doi.org/10.2514/6.2023-3736). [Online]. Available: <https://arc.aiaa.org/doi/10.2514/6.2023-3736>.
- [6] J. Anderson and B. Mary, *Introduction to Flight*, 9th. McGraw-Hill Higher Education, 2022, ISBN: 1260226743.

-
- [7] J. D. Anderson, *Hypersonic and High Temperature Gas Dynamics*, 2nd. American Institute of Aeronautics and Astronautics, 2006, p. 811, ISBN: 1563477807, 9781563477805.
 - [8] C. Vincenti and W. Kruger, *Introduction to Physical Gas Dynamics*. Krieger Pub Co., Jul. 1975, ISBN: 0882753096.
 - [9] K. J. Neitzel, "Thermochemical Modeling of Nonequilibrium Oxygen Flows," Ph.D. dissertation, The University of Michigan, 2017. [Online]. Available: https://deepblue.lib.umich.edu/bitstream/handle/2027.42/137049/kneitzel_1.pdf?sequence=1.
 - [10] N. M. Laurendeau, *Statistical Thermodynamics, Fundamentals and Applications*. Cambridge University Press, 2005, ISBN: 9780521846356.

Appendices